Real Fluid Dynamics

EXPERIMENT LAB REPORT

DISCALA-PORRO Julian Group B 2025-2026

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TP 1: Pitot tube velocity measurement

Introduction

The objective is to practice calibrating the pressure sensor using a water column manometer and to apply both the pitot tube and cobra probe for determining flow velocity and angular orientation. After completing these procedures, the relationship between velocity and yaw angle is represented graphically.

Experiment equipment

-Vertical wind tunnel

- Water column manometer

- A/D converter

- Pressure sensor

- Pitot tube

- Cobra probe

Calibrate pressure sensor with water column manometer

The sensor intended for subsequent measurements must first be calibrated using a water column manometer. This involves observing the variation in water level as a function of the motor's frequency output. To ensure accuracy, the bottom of the water bubble on the level indicator is aligned with a reference point marked by the spike tip.



Figure 2 Water column manometer



Figure 1 Level indicator



Figure 3 Vertical wind tunnel

Subsequently, the pressure difference is measured, and the corresponding height of the water column is recorded. The collected data are then used to construct a calibration curve, which exhibits a coefficient of determination (R^2) approaching unity, thereby confirming the reliability of the calibration process.

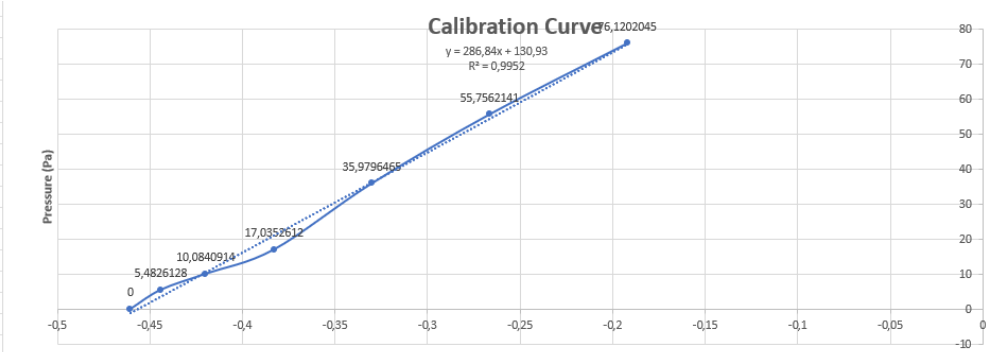
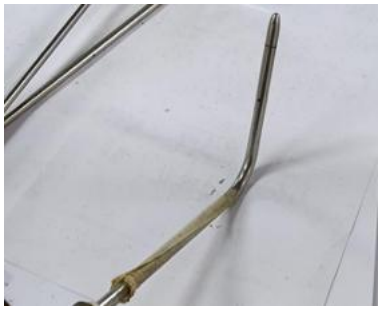


Figure 4 Pression plot

The pressure is expressed by the hydrostatic relation $P = \rho g h$. On this basis, the calibration of the sensor is considered complete.

Experiment with Pitot tube



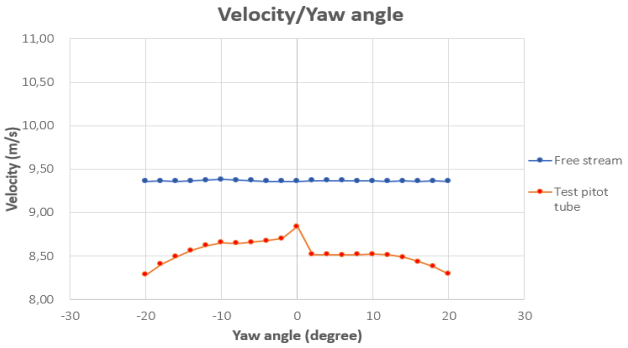
The Pitot tube is employed to establish the relationship between flow velocity and yaw angle. For this purpose, the test Pitot tube is connected to the pressure sensor to measure velocity, while the yaw angle is varied between $-20°$and $+20°$at a constant motor frequency of 30 Hz. The recorded measurements are organized in an Excel sheet. By applying the conservation principle together with Bernoulli's equation, the free-stream velocity is determined as

$$v_2 = \sqrt{\frac{2 \cdot (P_1 - P_2)}{\rho_{\text{air}} \left(1 - \frac{1}{6.76^2}\right)}}$$

Furthermore, using the dynamic pressure, the velocity measured by the Pitot tube is obtained as

$$v = \sqrt{\frac{2 \cdot P}{\rho_{\text{air}}}}$$

The resulting data are then represented graphically to illustrate the dependence of velocity on yaw angle.



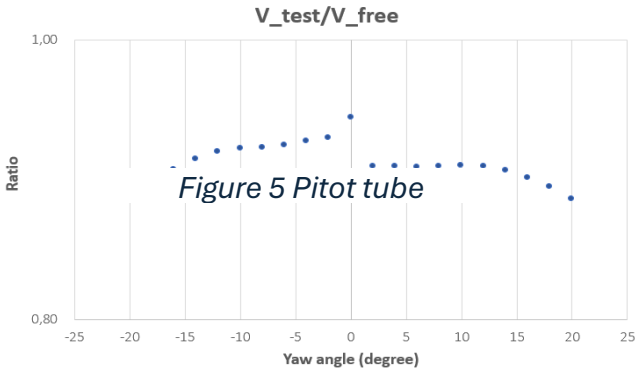


Figure 5 Pitot tube

Analysis

The first graph reveals an inconsistency in the experimental results: the curve corresponding to the Pitot tube test should coincide with the free-stream velocity. To analysis, the free-stream velocity is assumed to be 8.5 m/s. It is observed that when the yaw angle is below $-10°$or above $10°$, the velocity measured by the Pitot tube deviates significantly from the actual flow velocity. This indicates that the Pitot tube cannot be considered reliable for velocity determination outside the range of $-10°$to $+10°$. Moreover, the second graph demonstrates that the velocity recorded by the Pitot tube does not match the true free-stream velocity, a discrepancy attributable to imperfect sensor calibration.

Experiment with Cobra probe

In this part of the experiment, the Cobra probe is utilized by varying its yaw angle from $-20°$to $+20°$in increments of $2°$, while maintaining a constant motor frequency of 30 Hz. This procedure enables the determination of the relationship between pressure and yaw angle. Compared to the Pitot tube, the Cobra probe offers enhanced capabilities, as it can simultaneously measure both velocity and flow direction across different angular positions. The resulting data are presented in the graph illustrating the correlation between pressure and yaw angle.

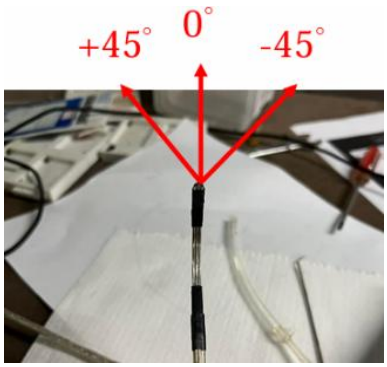


Figure 6 Cobra probe

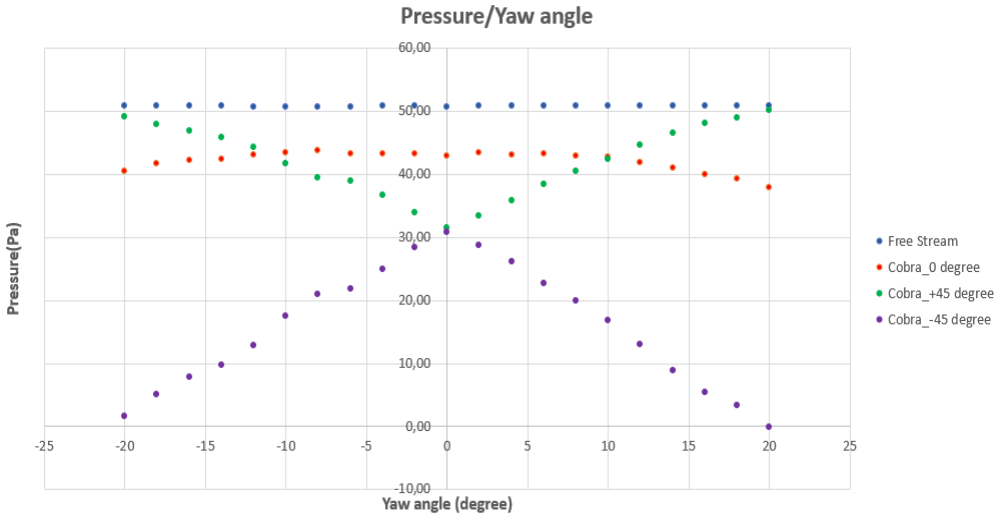


Figure 7 Pressure as a function of angle

Analysis

The Cobra probe exhibits symmetry with respect to the $0°$axis, as evidenced by the fact that the curves corresponding to $+45°$and $-45°$are mirror images about this axis. At larger yaw angles, however, the stagnation point may be lost, causing the measured pressure to converge toward the ambient static pressure of the free stream. This phenomenon is observed in the graph when the yaw angle exceeds $|\ 20°\ |$.

To enhance the accuracy of the results, the following data associations are introduced:

p_0 =Cobra probe pressure at $0°$

p_1 =Cobra probe pressure at $-45°$

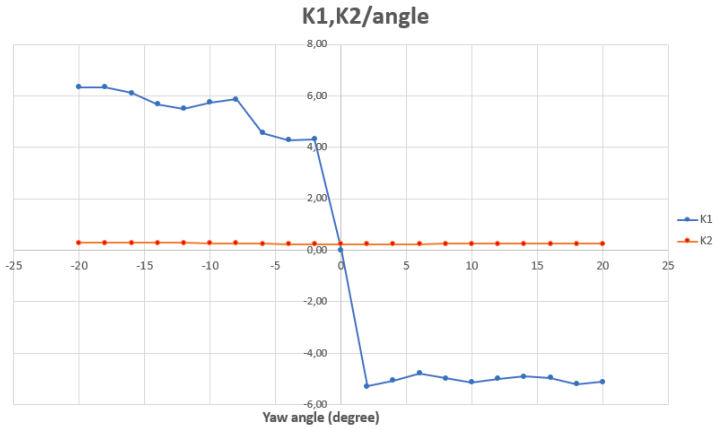
p_2 =Cobra probe pressure at $+45°$

Using these values and the yaw angle β, the correction coefficients $K_1$and $K_2$are defined as:

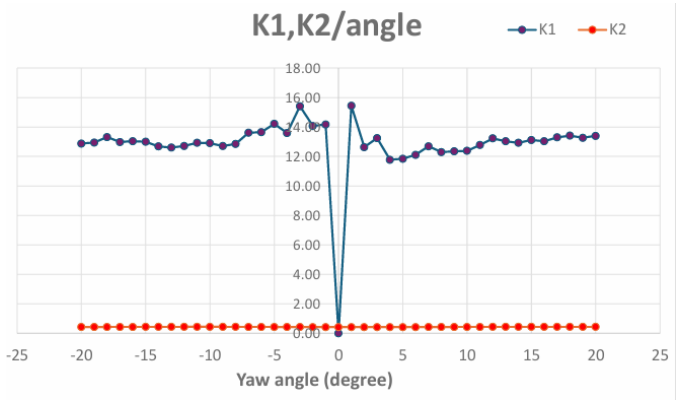
$$K_1 = \beta \cdot \frac{p_0 - \bar{p}}{p_1 - p_2}, \text{where } \bar{p} = \frac{p_1 + p_2}{2}$$

$$K_2 = \frac{p_0 - \bar{p}}{0.5 \cdot \rho_{\text{air}} \cdot V^2_{\text{free-stream}}} = \frac{p_0 - \bar{p}}{P_{\text{free-stream}}}$$

These correction coefficients allow for improved calibration and more reliable interpretation of the Cobra probe measurements.



Experiment resut



Experiment resut

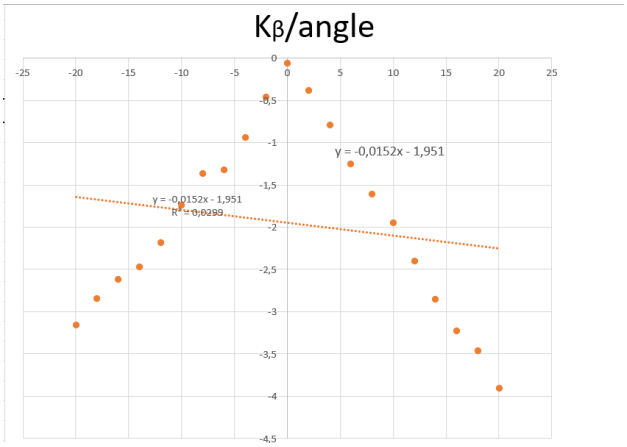
The left-hand graph illustrates the variation of the coefficients K_1 and K_2 as a function of the yaw angle. During the experiment, a technical issue occurred due to improperly connected wires, necessitating a repetition of the procedure. Nevertheless, residual experimental uncertainties remained, and the results obtained (right-hand graph) were close to expectations but not fully consistent.

The coefficient K_2 remains essentially constant with respect to the yaw angle, indicating that the normalized term is largely independent of angular variation. This behavior serves as a reliable indicator of calibration quality for the measurement of average pressures. In contrast, K_1 is generally stable across most of the angular range but diverges near yaw $\approx 0°$. This divergence arises from the definition of K_1, since the symmetry of the measurements p_1 and p_2 about $0°$ causes the difference $(p_1 - p_2)$ to vanish, rendering K_1 ill-conditioned. Consequently, the values of K_1 are not reliable in the vicinity of $0°$.

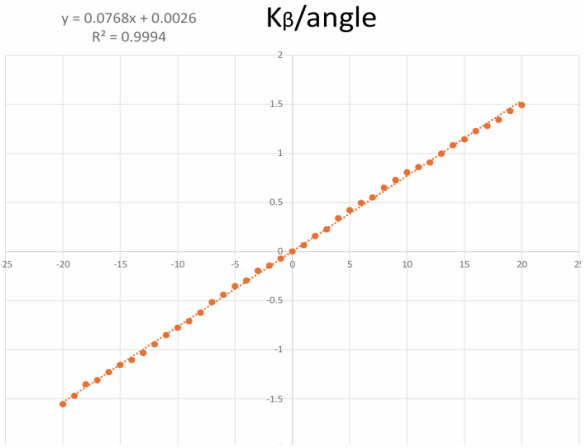
Following this, the direction coefficient is introduced as

$$K_\beta = \frac{p_1 - p_2}{p_0 - \bar{p}}, \text{where } \bar{p} = \frac{p_1 + p_2}{2}.$$

By dividing this direction coefficient by the yaw angle, the resulting relationship is presented in the left-hand graph.



Experiment result



Expected result

The relationship between K_β and the yaw angle is expected to be linear. Like the K_1 and K_2/anglegraphs, certain experimental issues such as connection errors or measurement uncertainties were encountered. Nevertheless, the observed trend is approximately linear and symmetric with respect to the y-axis. Ideally, a single straight line passing through the origin should be obtained.

Upon analyzing the expected outcome, the data reveal a linear curve that nearly intersects the origin, with a coefficient of determination $R^2 = 0.9994$, which is very close to unity. The coefficient K_β characterizes the variation of lateral pressure $(p_1 - p_2)$ relative to the axial variation $(p_0 - \bar{p})$. This result confirms that the Cobra probe is properly calibrated and that the recorded values are reliable.

Conclusion

In this experiment, the procedures for sensor calibration were successfully carried out, and the distinctions between the Pitot tube and the Cobra probe were examined. The Pitot tube proved to be a simple and reliable instrument, particularly well-suited for measuring flows aligned with a straight axis. In contrast, the Cobra probe demonstrated a clear advantage in the analysis of more complex flow structures, as it is capable of simultaneously measuring both velocity and flow direction.

TP 2: Flow visualization in water channel

Introduction

The main objective was to analyze flow separation for different geometries: a circular cylinder, a slotted cylinder, and a NACA 0012 airfoil subjected to various angles of attack. By adjusting the water flow rate and recording the flow at high frame rates, we were able to compare the flow fields between different geometries, and study how the separation points shift depending on the geometry and experimental conditions. An additional goal of the experiment was to estimate the Strouhal number for the different flow cases based on the measured vortex shedding frequencies.

Experiment equipment

- Watter Channel
- 15mm circular cylinder
- NACA 0012 model
- 15mm slotted circular cylinder



Figure 9 NACA 0012 & Cylinder

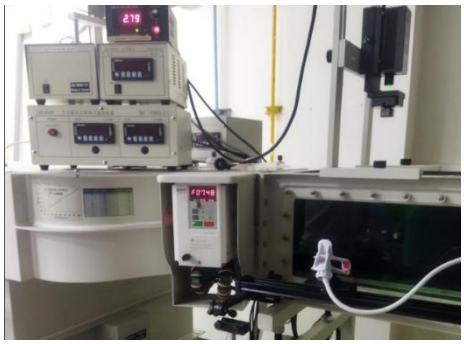


Figure 8 Watter channel

Cylinders

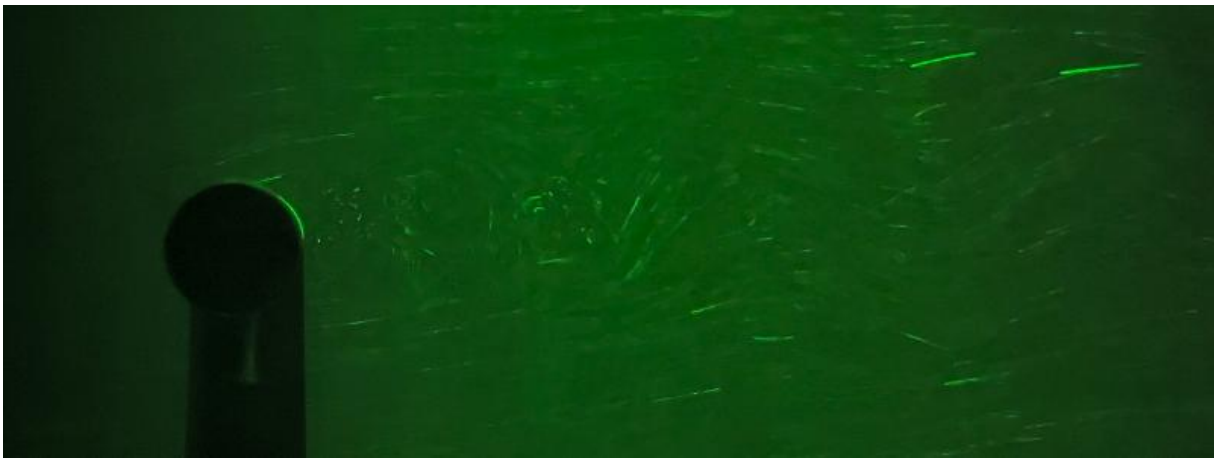


Figure 10 Circular cylinder flow visualization

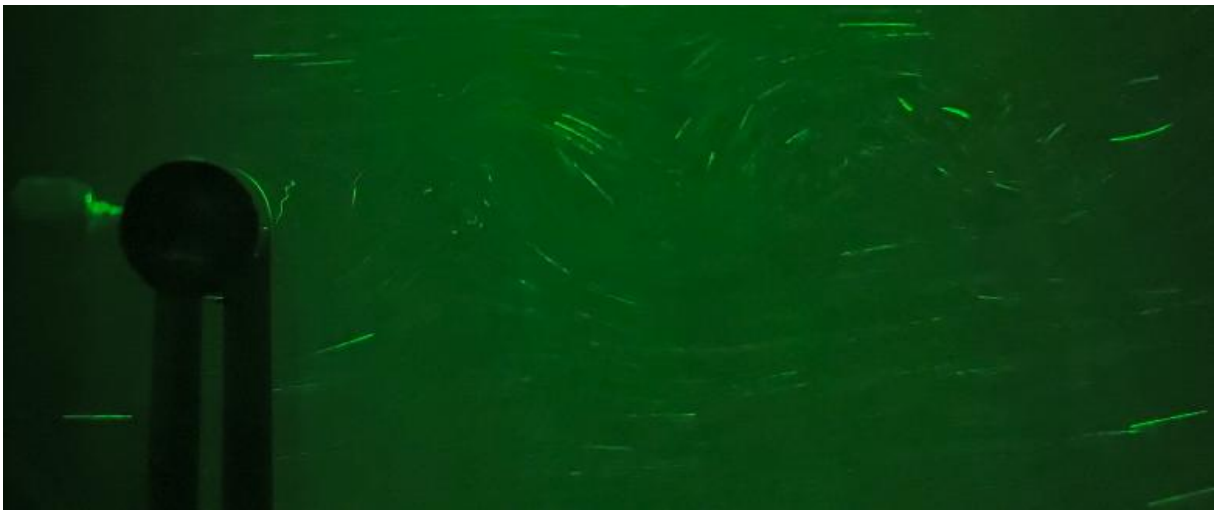


Figure 11 Slotted circular cylinder (β= 0) flow visualization



Figure 12 Slotted circular cylinder (β= 90°) flow visualization

Objectives

In this experiment, we studied the vortex-shedding process behind a circular cylinder placed in a water channel and analyzed how the Reynolds number and Strouhal number evolve with flow conditions. The main objective was to observe the formation of the Von Kármán vortex and quantify the characteristic shedding frequency.

The experimental setup consisted of a straight water channel with a controllable flow rate. The flow speed was adjusted by selecting different forcing frequencies on the control panel of the channel. Five flow conditions were tested, corresponding to forcing frequencies of 7.5 Hz, 10.5 Hz, 13.5 Hz, 16.5 Hz, and 19.5 Hz. For each case, we recorded the wake downstream of the cylinder.

We used Particle Image Velocimetry (PIV) to extract the velocity field from the videos. This involved seeding the water with tracer particles, illuminating the

measurement section with a light sheet and performing cross-correlation analysis using the PIVlab toolbox in MATLAB. From the processed velocity fields, we obtained the mean flow velocity U for each forcing frequency.

Frequency (Hz)	Uinf (m/s)
7,5	0,0987
9	0,1182
10,5	0,1377
12	0,1572
13,5	0,1767
15	0,1962
16,5	0,2157
18	0,2352
19,5	0,2547

The vortex-shedding frequency $f$was determined by analyzing the temporal variation of velocity fluctuations in the wake or by counting repeated vortex structures.

Despite having video data only for a subset of the possible forcing frequencies, we were able to extract consistent flow characteristics across all available cases. The results show that the Strouhal number remains nearly constant as the Reynolds number increases, aligning with the classical value of $St \approx 0.2$for laminar vortex shedding behind a circular cylinder. Small deviations can be attributed to experimental uncertainties such as image resolution, limited sampling frequencies, PIV noise, and incomplete frequency coverage.

Calcul of vortex shedding frequency

The objective of this part is to plot the Strouhal number graph as a function of the Reynolds number.

We have: $St = \frac{f*D}{U_\infty}$

Where f is the vortex shedding frequency: $f = \frac{wake\ number}{time\ (sec)}$

D the diameter of the cylinder $D = 0.015\ m$

U_∞ the fluid speed: $U_\infty = 0.013 * freq + 0.0012$

$freq$ is the frequency of the water channel

We also have: $Re = \frac{\rho * U_\infty * D}{\mu}$

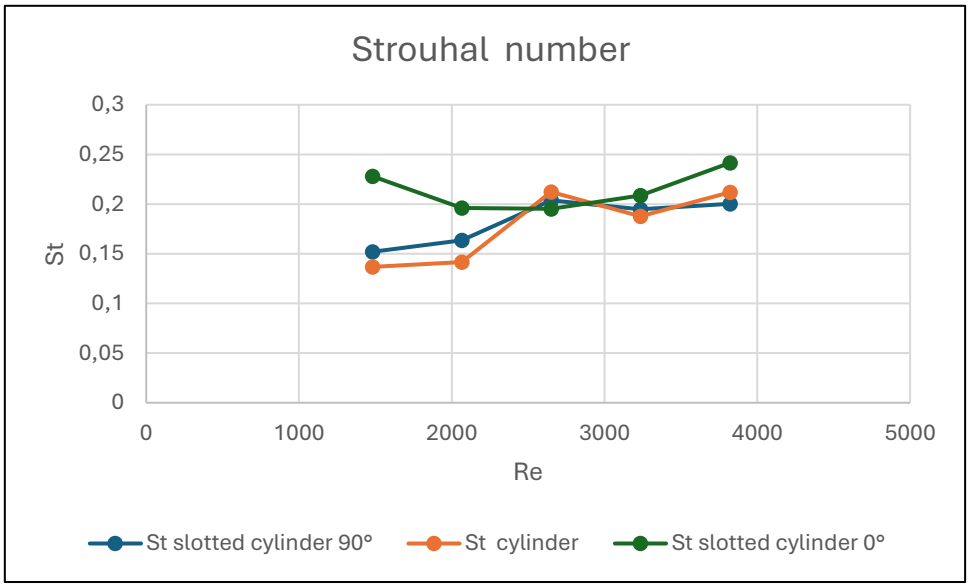
Where $\rho = 1000\ \mathrm{Kg}/m^3$

$\mu = 1 * 10^{-3} Pa.s$

We slowed down the video and manually counted the wake number to determine the shedding frequency.

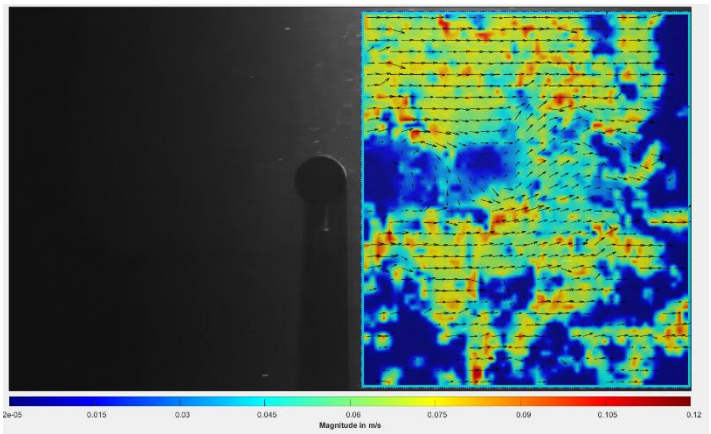
Thus, we have the following table for wake number and graph for Strouhal number:

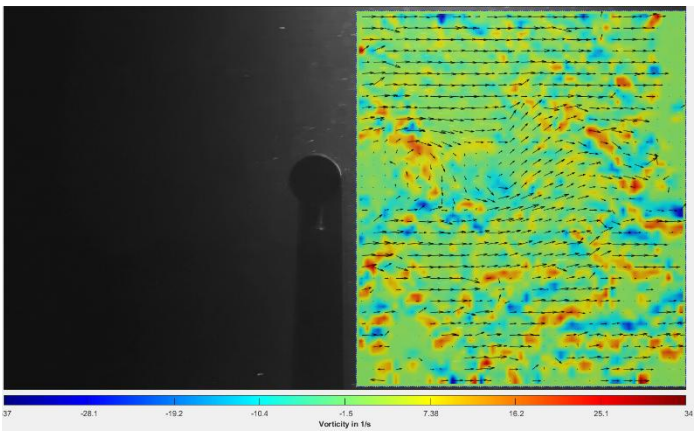
Vortex shedding frequency	7.5	10.5	13.5	16.5	19.5
Circular Cylinder	1.0	1.5	2.4	2.8	3.4
Slotted Circulat Cylinder 0°	0.9	1.3	2.5	2.7	3.6
Slotted Circulat Cylinder 90°	1.5	1.8	2.3	3.0	4.1



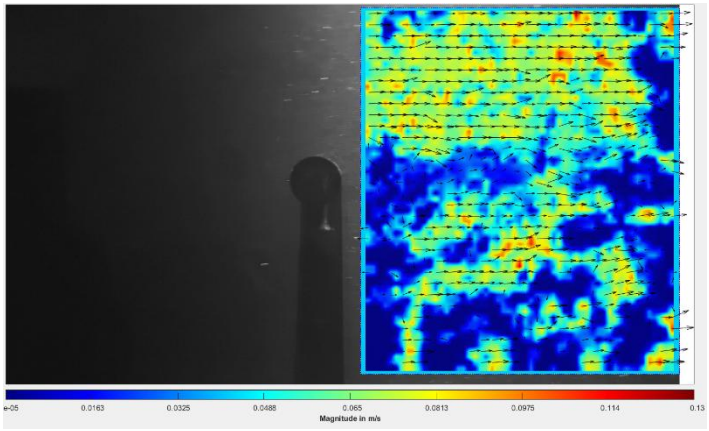
PIV Result cylinder

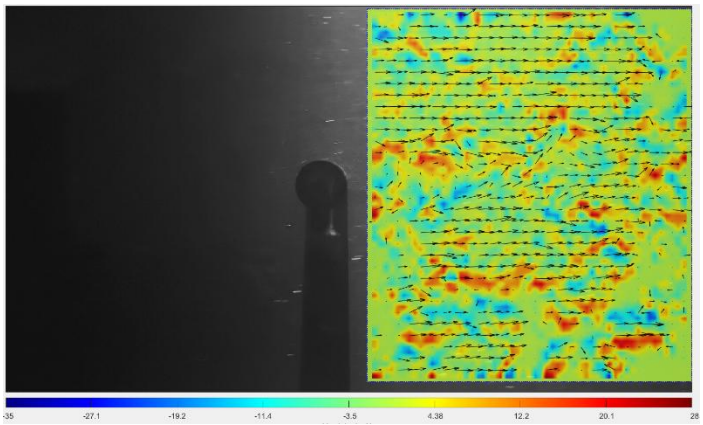
PIV Result Circular cylinder 7.5Hz



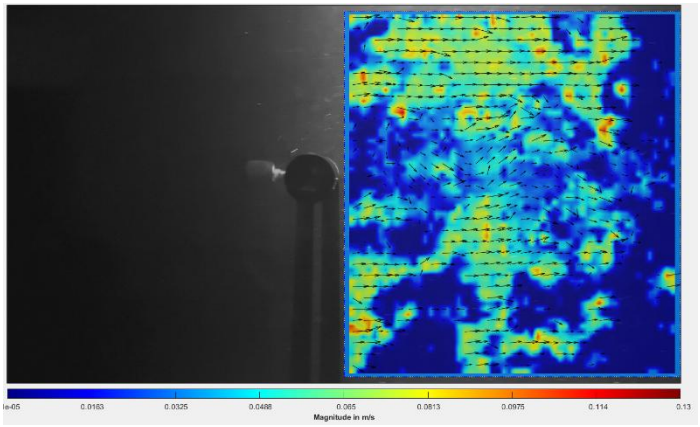


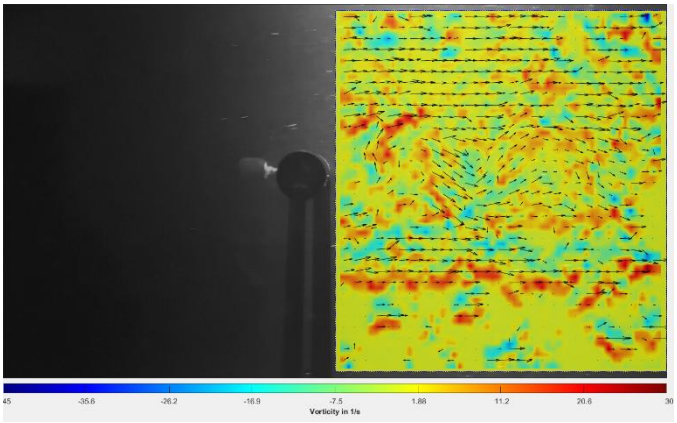
PIV Result (Slotted circular cylinder β= 0°) 7.5Hz

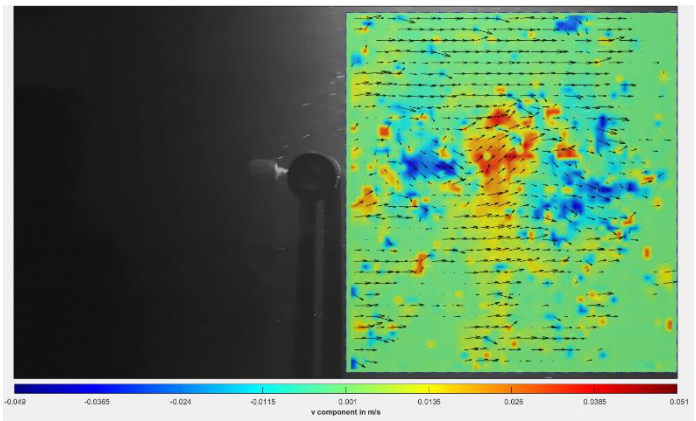


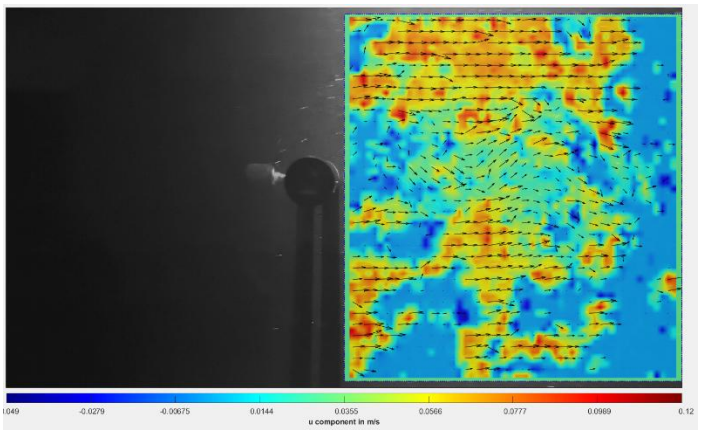


PIV Result (Slotted circular cylinder β= 90°) 7.5Hz



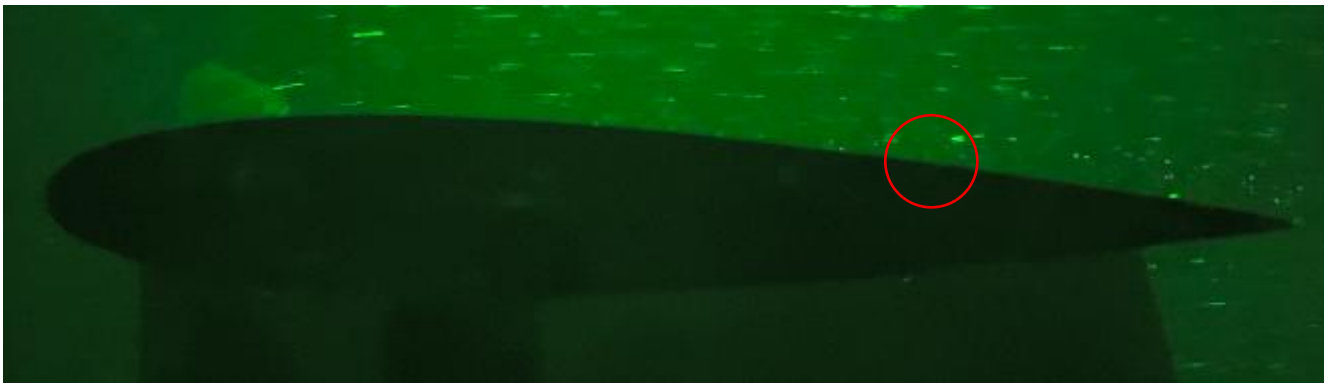


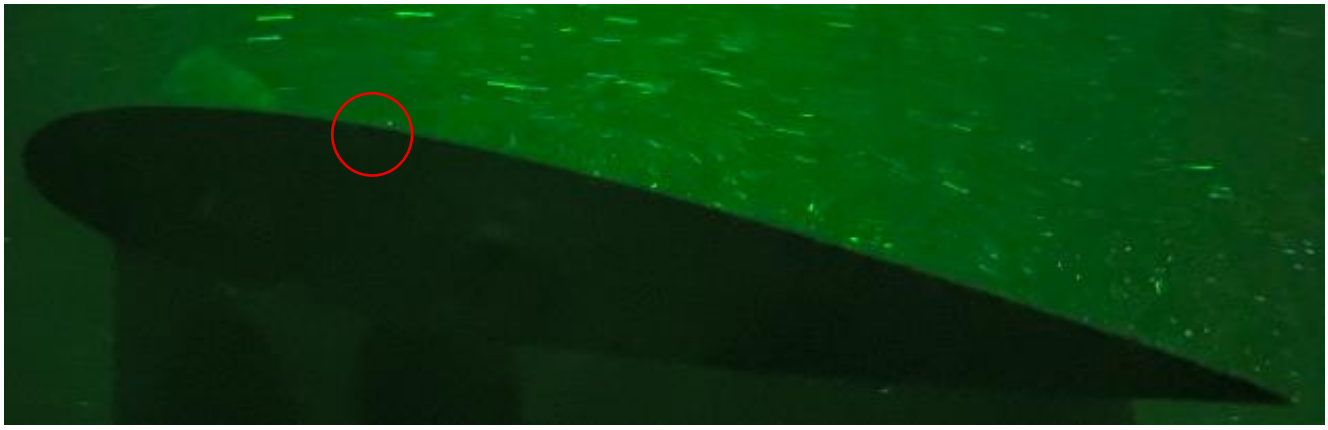


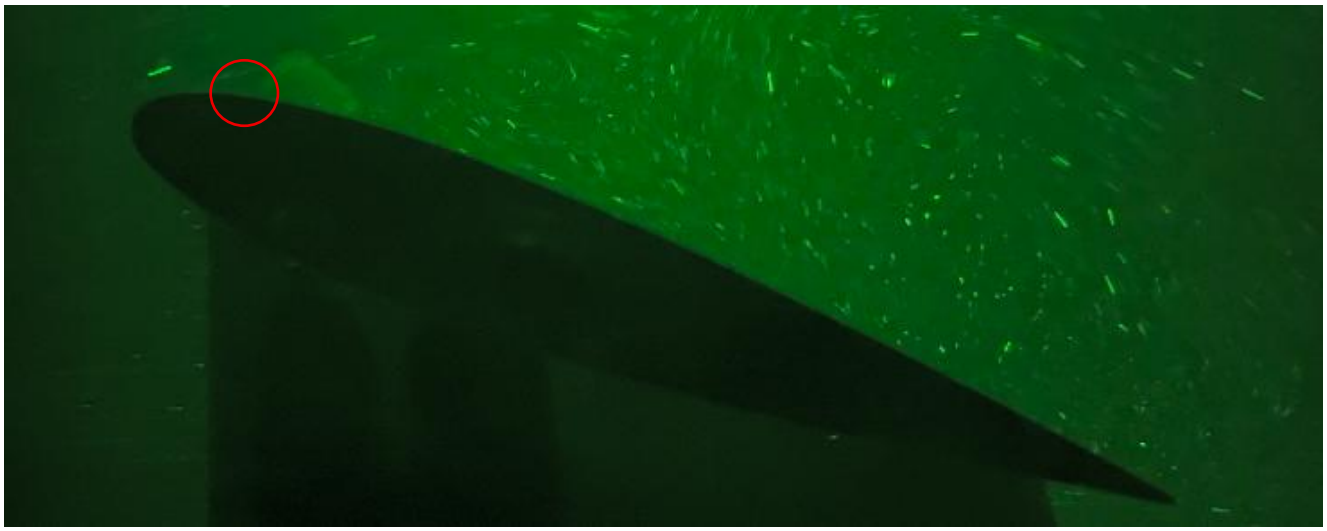


PIV Result NACA 0012

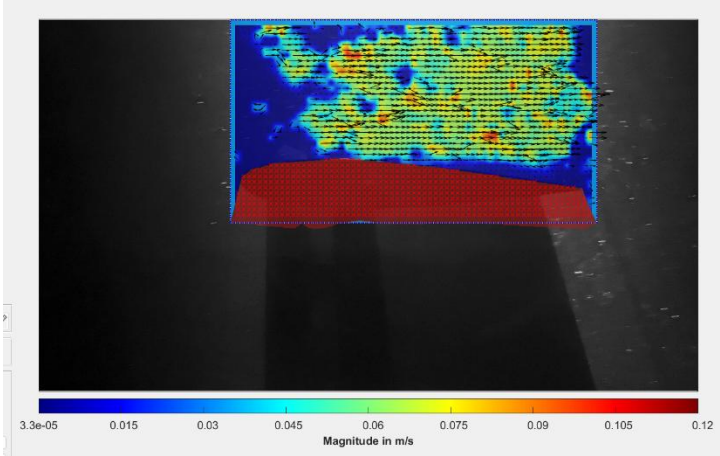
The red circles below represent separation point. The separation point was identified from the PIV data by examining the tangential velocity along the airfoil surface and locating the position where the velocity dropped to zero and reversed sign, indicating boundary-layer detachment. The PIV results also showed that the turbulent separated region becomes larger as the angle of attack increases.

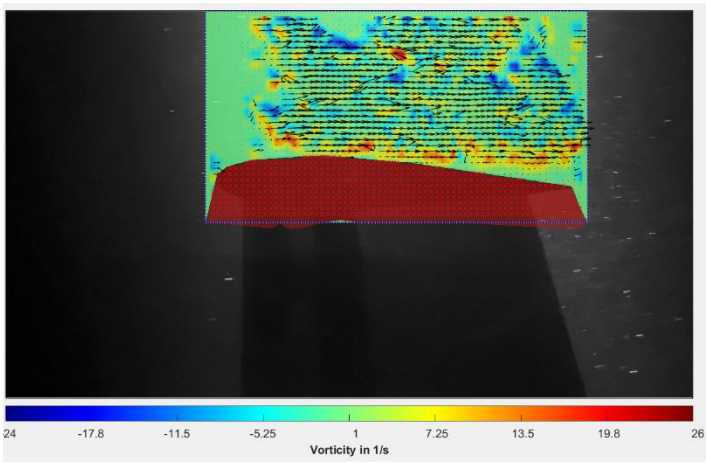




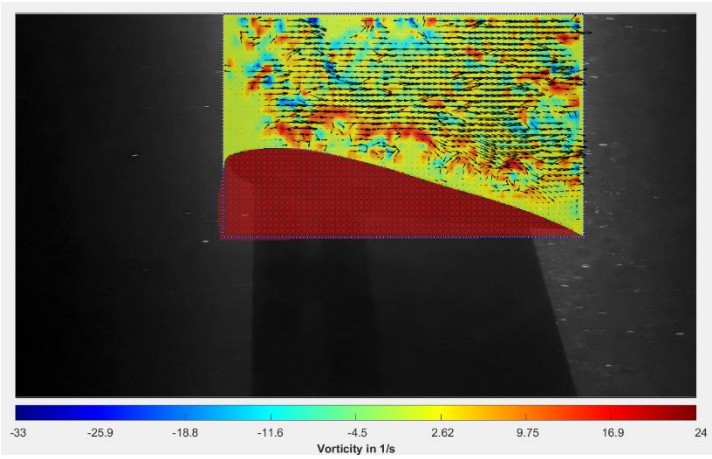


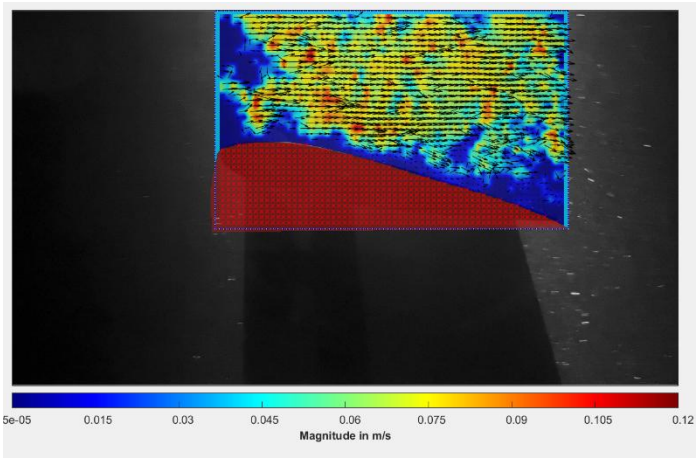
PIV Result (NACA 0012 PIV) AOA=0°



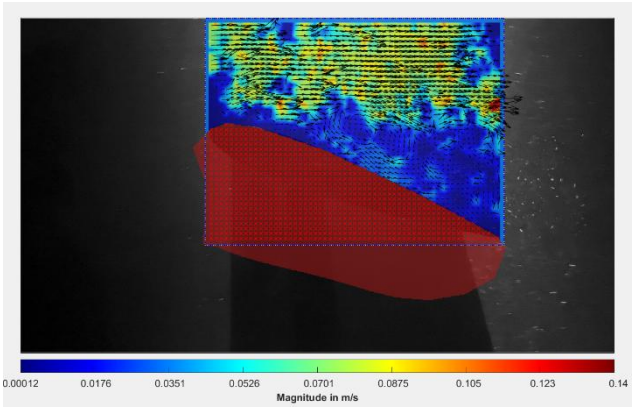


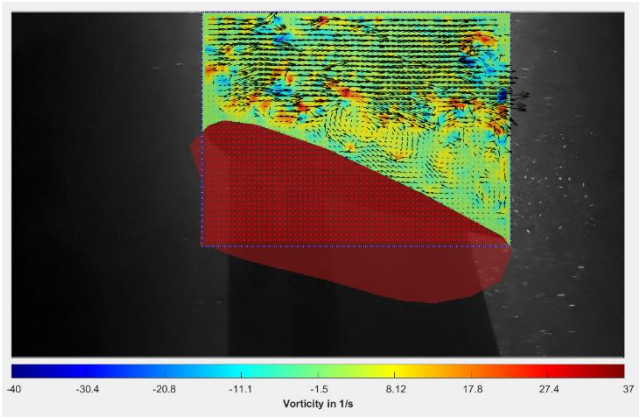
PIV Result (NACA 0012 PIV) AOA=10°





PIV Result (NACA 0012 PIV) AOA=20°





On the PIV vorticity fields, we observe strong vorticity concentrations around the leading edge, a thin shear layer along the suction side, and a separated turbulent wake behind the airfoil. As the angle of attack increases, the separation region and turbulent structures become larger and more chaotic.

Conclusion

In this experiment, we investigated the relationship between the Reynolds number, the Strouhal number, and vortex shedding behind a circular cylinder in a water channel. By varying the forcing frequency of the flow and measuring the corresponding flow velocity using PIV, we were able to estimate both the vortex-shedding frequency and the resulting nondimensional coefficients.

The results confirm the classical behavior of cylinder wake dynamics. As the Reynolds number increases, the Strouhal number remains nearly constant over a broad range, consistent with the well-established value of approximately $St \approx 0.2$ for laminar vortex shedding. Although some deviations were observed likely due to experimental limitations such as incomplete frequency sampling, camera resolution, and flow-rate calibration, the overall trend aligns with theoretical and empirical data found in the literature.

TP 3: Wake flow experiment in low-speed wind tunnel

Introduction

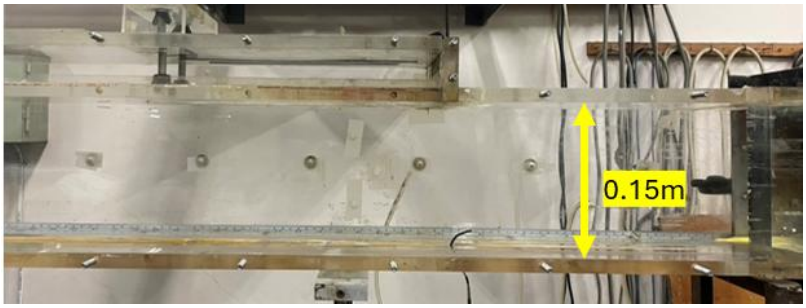
This experiment is designed to investigate the wake of a circular cylinder within a closed-return, low-speed wind tunnel. The wake generated behind a bluff body such as a circular cylinder represents a classical problem in fluid mechanics, as it encompasses complex phenomena including flow separation, turbulence production, and periodic vortex shedding. These mechanisms are of considerable importance in a wide range of engineering applications, notably in aerodynamics, civil engineering, and mechanical design.

The primary objective of the study is to examine the mean velocity distribution, turbulence intensity, and Strouhal number in the cylinder's wake. Measurements are obtained using a pressure transducer, a Pitot tube, and a single-wire hot-wire probe connected to a constant-temperature anemometry system and an A/D converter. This instrumentation enables accurate determination of both the mean flow characteristics and the unsteady velocity fluctuations.

The velocity and frequency data collected during the experiment are subsequently processed to derive nondimensional velocity, turbulence intensity, and Strouhal number under controlled conditions. These parameters provide a detailed characterization of the wake structure, facilitate identification of the periodic vortex-shedding regime, and allow for comparison of the experimental findings with theoretical predictions and reference data available in the literature.

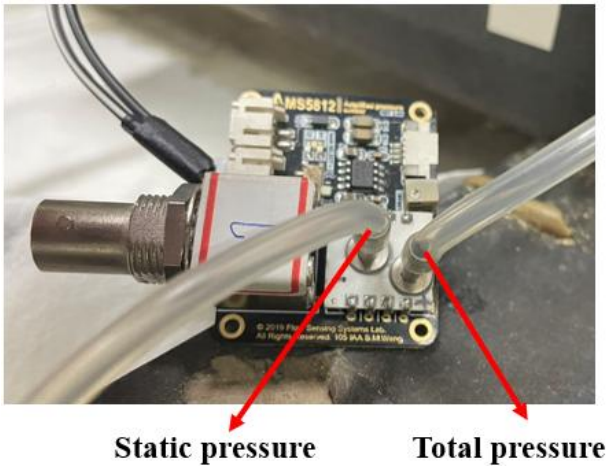
Equipment

Closed-return wind tunnel



The closed-return wind tunnel is employed to produce a controlled, steady, and repeatable airflow around the cylinder. The air circulates within a closed loop driven by a fan, with the velocity adjustable in the range of 2to 22 m/s. The test section, characterized by a cross-sectional area of $0.15\ \text{m} \times 0.15\ \text{m}$, provides well-defined aerodynamic conditions. Its significance in this experiment lies in ensuring a stable and uniform incoming flow, which is essential for accurately investigating the wake formation behind the cylinder and for obtaining reliable and reproducible measurements of velocity and turbulence.

Pressure transducer (AMS-5812



The AMS-5812 pressure transducer is employed to measure both static and total pressure within the flow field. Its operating principle consists of converting pressure variations into an electrical voltage signal ranging from 0.5 to 4.5 V. The primary function of this device is to ensure precise pressure measurements. In the context of the present experiment, its importance lies in enabling the determination of dynamic pressure and, consequently, the mean flow velocity. This parameter serves as a fundamental reference for the calculation of both the Reynolds number and the Strouhal number.

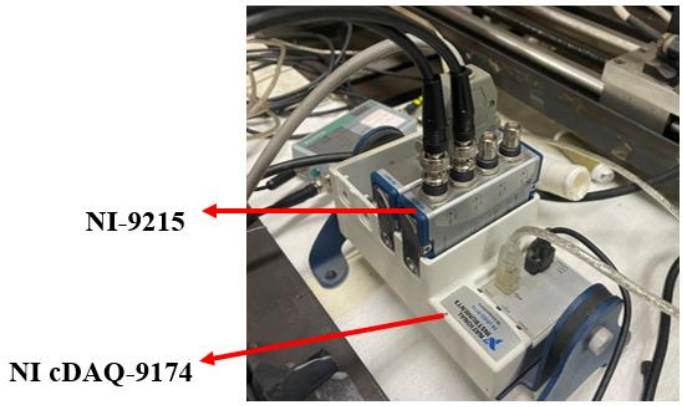
Constant Temperature Anemometry (CTA)

T



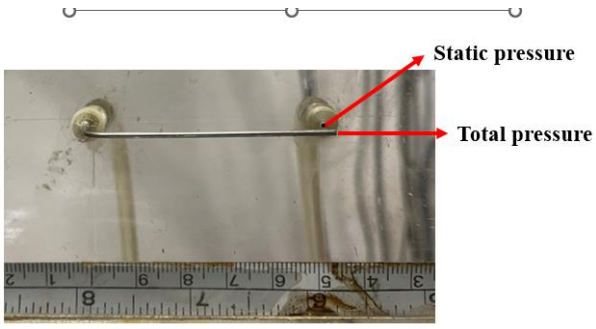
The constant-temperature anemometry (CTA) system is employed to capture rapid velocity fluctuations in the flow. Its operating principle relies on a fine heated wire whose temperature is maintained constant by continuously adjusting the electrical current as the surrounding airflow cools it. The magnitude of the current variation is directly correlated with the flow velocity. The primary function of the CTA is to measure unsteady velocity components and turbulence. Within the framework of this experiment, its importance lies in enabling a precise characterization of the turbulent structures in the cylinder wake and in facilitating the detection of vortex-shedding phenomena.

A/D converter



The A/D converter system, consisting of the NI cDAQ-9174 chassis and the NI-9215 module, is utilized to transform analog voltage signals from the sensors into digital data that can be processed by a computer. The device operates within an input range of ± 10 Vand provides a maximum sampling rate of 100 kS/sper channel. Its primary function is to enable high-speed data acquisition. In the context of this experiment, its importance lies in the ability to record rapid pressure and velocity fluctuations with high temporal resolution, a requirement that is essential for conducting frequency analysis and for the accurate determination of the Strouhal number.

Pitot tube



The Pitot tube is employed to determine the mean flow velocity. Its operating principle relies on measuring the total pressure in the direction of the flow and the static pressure laterally. The difference between these two pressures corresponds to the dynamic pressure, from which the velocity can be derived using Bernoulli's equation. The primary function of the Pitot tube is to provide a reliable assessment of the average flow speed. Within the framework of this experiment, its significance lies in serving as a reference instrument to validate the velocity obtained from the hot-wire system and to ensure the accuracy of the experimental conditions.

Single-wire probe



The single-wire probe is employed to measure the instantaneous local velocity of the airflow. Its operating principle is based on a fine heated wire whose cooling rate is directly influenced by the local flow speed. When positioned within the wake of the

cylinder, the probe can detect rapid velocity variations. Its primary function is to capture local velocity fluctuations. In the context of this experiment, its importance lies in enabling a detailed analysis of turbulence intensity and vortex-shedding phenomena across different regions downstream of the cylinder, which is fundamental to understanding the dynamics of wake behavior.

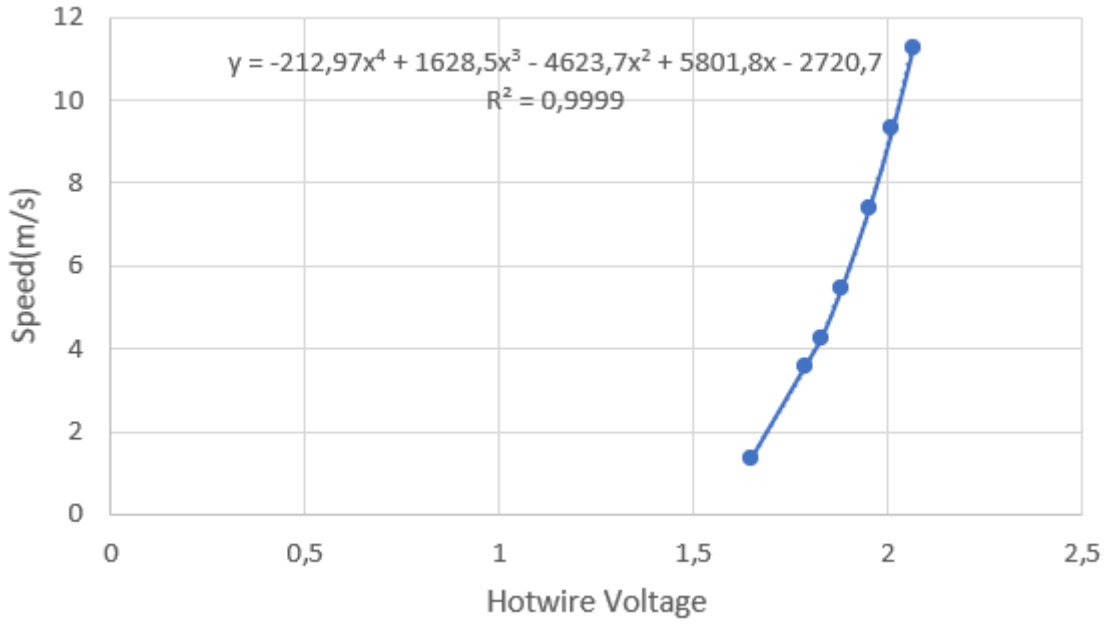
Calibration and Experimentation:

During the calibration phase, the objective is to establish a precise correlation between the electrical signal delivered by the hot-wire probe and the actual air velocity measured by the Pitot tube. This step is indispensable, as the hot-wire anemometer does not directly measure velocity but rather records a voltage that depends on the cooling rate of the heated wire exposed to the airflow. To employ the hot-wire probe for accurate velocity and turbulence measurements, it is therefore necessary to convert the voltage into a physical velocity through a calibration law.

The calibration procedure is performed by positioning both the Pitot tube and the hot-wire probe within the uniform flow of the wind tunnel, where the velocity is steady and well controlled. The Pitot tube serves as the reference instrument, providing reliable measurements of the mean air velocity based on Bernoulli's principle. By recording the total and static pressures, the dynamic pressure is obtained, from which the air velocity is calculated. Simultaneously, the hot-wire probe measures a voltage corresponding to the same flow conditions.

This process is repeated across several operating points to generate paired values linking hot-wire voltage to air velocity. Specifically, the excitation frequency of the wind tunnel control system was progressively varied to 5, 10, 12, 15, 20, 25, and 30 Hz, each setting producing a different flow speed. At every step, the software recorded both the Pitot tube measurements and the hot-wire voltage. The acquisition system collected voltage signals as a function of time and frequency, enabling the extraction of stable mean values for each operating condition.

This systematic approach ensured that the calibration encompassed a sufficiently broad velocity range, thereby guaranteeing reliability across the entire experimental domain. Once all data were collected, the air velocity was calculated from the Pitot tube measurements for each frequency and associated with the corresponding hot-wire voltages. The resulting calibration curve, representing airflow velocity as a function of hot-wire voltage, exhibits a strongly non-linear relationship characteristic of hot-wire anemometry. The data points follow a smooth increasing trend, confirming that higher voltages correspond to higher air velocities.



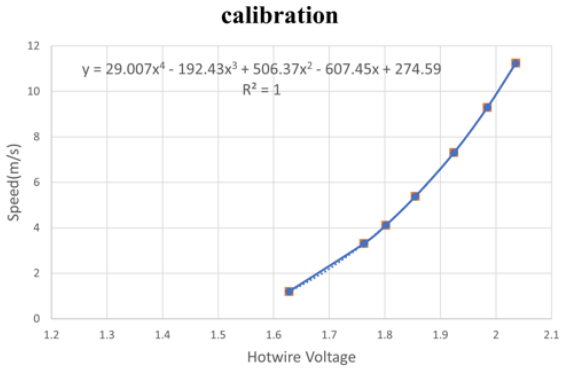
A polynomial regression was applied to the experimental calibration points to establish an analytical calibration law. The resulting equation is of fourth order, which reflects the complex thermal behavior of the hot-wire sensor. The coefficient of determination was found to be $R^2 = 0.9999$, indicating an excellent agreement between the experimental data and the regression model. This very high value confirms the accuracy of the calibration and validates the use of the polynomial law for converting voltage signals into velocity values throughout the remainder of the experiment.

The shape of the calibration curve demonstrates that the sensitivity of the hot-wire probe increases with velocity. At low voltages, the velocity rises gradually, whereas at higher voltages, the velocity increases more rapidly. This behavior is consistent with the physics of convective heat transfer around the wire, where the cooling effect of the airflow intensifies as velocity increases. Once the calibration curve was established, all hot-wire voltage measurements obtained in the wake of the cylinder were converted into instantaneous velocity values using the polynomial equation. This conversion was a fundamental step, as it enabled the computation of mean velocity, turbulence intensity, and the dominant frequency of velocity fluctuations. These quantities were subsequently used to determine key aerodynamic parameters such as the Strouhal number and to analyze the wake structure behind the cylinder.

In summary, the calibration phase ensured the reliability of the experimental measurements. By systematically varying the excitation frequency, recording both Pitot and hot-wire data, calculating the air velocity, constructing the calibration curve, and validating it with a high R^2value, the accuracy and physical consistency of all subsequent flow measurements were guaranteed.

Finally, comparison with the reference calibration curve provided in the course material revealed the same global behavior. In both cases, the airspeed increases non-linearly with hot-wire voltage, and the curve exhibits a convex shape with a stronger rise at higher voltages. The fourth-order polynomial fit, and the near-unity coefficient of determination confirms that the experimental data follows the expected trend and lies within a similar velocity range. Minor discrepancies can be attributed to experimental uncertainties such as sensor positioning or temperature variations, but overall, the calibration is consistent and physically coherent with the reference results.

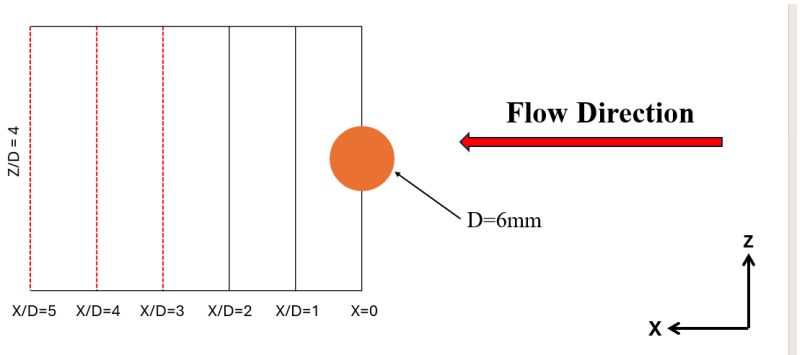
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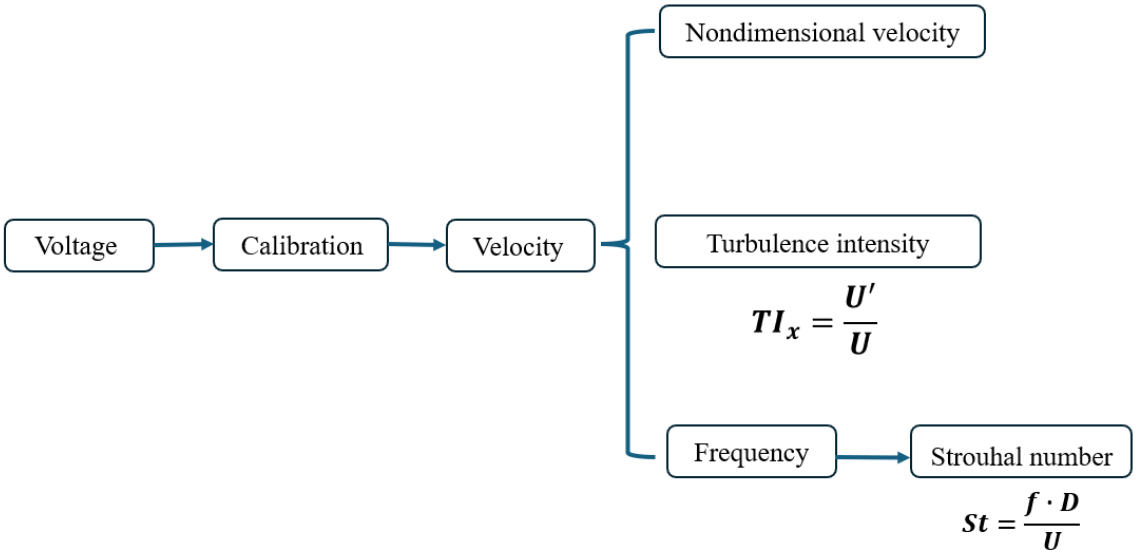
Experiment method and result

The experiment is conducted in a closed-return wind tunnel to investigate the wake generated by a circular cylinder subjected to a uniform airflow. The flow direction is aligned with the x-axis, while measurements are performed in the downstream region along the z-axis. This configuration provides controlled and repeatable conditions for examining the flow field behind the cylinder. The use of a closed-loop wind tunnel ensures stable upstream flow, which is essential for obtaining reliable aerodynamic measurements.

The test cylinder has a diameter $D = 6$ mm and is positioned perpendicular to the incoming flow. Measurements in the wake are carried out at three downstream locations corresponding to the non-dimensional distances $X/D = 3$, 4, and 5. At each of these sections, the velocity field is investigated across the transverse direction to characterize the structure of the wake and to analyze its evolution with increasing distance from the cylinder.



The scanning technique consists of moving the measurement probe incrementally across the wake at each X/D position. By systematically sweeping the measurement zone in the transverse direction, it becomes possible to reconstruct velocity profiles and turbulence intensity distributions, thereby identifying the principal characteristics of the wake, such as the velocity deficit and regions of elevated turbulence. This approach provides a spatial characterization of the wake structure rather than a simple local measurement, offering a more comprehensive description of the flow field downstream of the cylinder.



The data processing methodology is summarized in the diagram. The voltage signal measured by the hot-wire probe is first converted into velocity using the previously established calibration law. From this velocity signal, the non-dimensional velocity is computed, the turbulence intensity is determined as the ratio between velocity fluctuations and the mean velocity, and the dominant frequency is extracted to calculate the Strouhal number. This processing chain enables the derivation of all principal aerodynamic parameters of the wake directly from the raw electrical signal.

The experiment is conducted under well-defined and controlled conditions. The free-stream excitation frequency is set to 20 Hz, corresponding to a free-stream velocity of 7.6 m/s. The ambient temperature during the measurements is 28°C, and the kinematic viscosity of air is taken as 1.59×10^{-5} m^2/s. At each measurement location, 75 samples are recorded to ensure statistical reliability. Experimentally, a continuous acquisition lasting 10 minutes is performed at each position. This extended acquisition time provides a sufficiently large dataset to accurately determine the mean velocity, turbulence intensity, and dominant frequency of the flow. These quantities are subsequently used to analyze the wake behavior and to compute the Strouhal number along with other non-dimensional parameters.

Matlab code and result

The processing of the measurements was performed using a MATLAB script that consolidates all information recorded during the experiment. Two Excel spreadsheets prepared beforehand serve as inputs: the first contains the calibration results, including the polynomial relation between hot-wire voltage and velocity, while the second stores the geometric information, specifically the Z/D positions corresponding to the different X/D sections.

The calibration law used to convert hot-wire voltage into velocity is expressed as a fourth-order polynomial derived from the experimental curve:

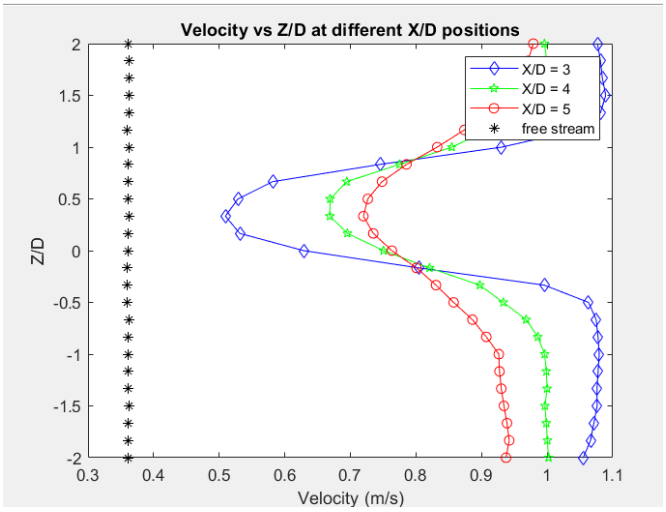
$$y = -212.97x^4 + 1628.5x^3 - 4623.7x^2 + 5801.8x - 2720.7$$

In addition, a second calibration provides the relation between pressure and voltage, expressed as a linear law:

$$y = 259.329x - 15.0411$$

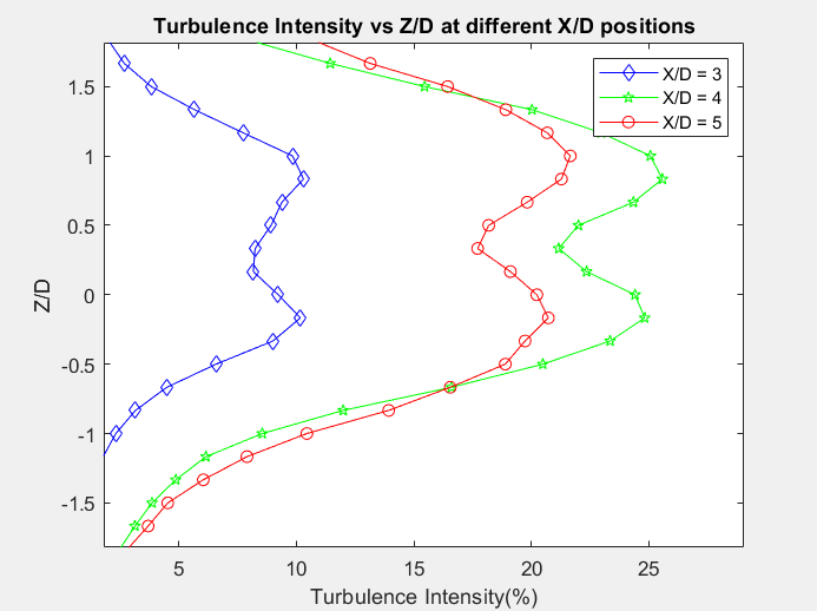
The program reads the Z/D coordinates and automatically scans all text files exported by the acquisition software, each file corresponding to a single measurement point in the wake. For every file, the script extracts the hot-wire and reference voltages, converts them into velocity using the calibration equations, and computes both the mean velocity and the velocity fluctuations. From these time series, a Fourier transform is applied to obtain the frequency spectrum and to identify the dominant vortex-shedding frequency.

All processed data are then organized by X/D section to calculate the non-dimensional velocity, turbulence intensity, and Strouhal number. Finally, the script generates the four graphs that form the basis of the subsequent analysis.



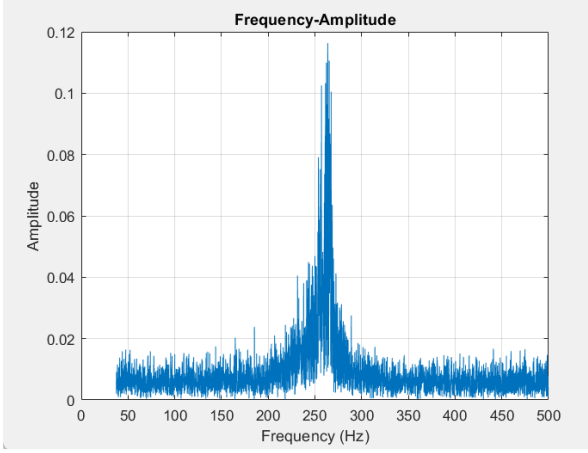
The first graph presents the velocity profiles as a function of $Z/D$for different downstream positions defined by $X/D = 3,4,$and 5. It illustrates the evolution of the mean velocity across the wake at these three locations. A pronounced velocity deficit is observed near $Z/D = 0$, corresponding to the wake region generated by the cylinder. As the downstream distance increases from $X/D = 3$to $X/D = 5$, the profiles become progressively smoother and the minimum velocity rises, indicating that the wake gradually recovers toward the free-stream velocity.

This behavior is consistent with classical wake theory, as viscous diffusion and turbulent mixing act to diminish the velocity deficit with increasing downstream distance. Furthermore, the tendency of the profiles to converge toward the free-stream value at large | Z/D |confirms that the external flow has been correctly captured by the measurements.



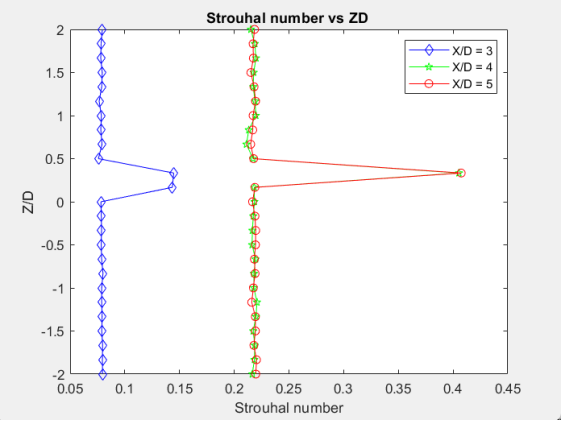
The second graph depicts the turbulence intensity as a function of $Z/D$for different downstream positions defined by $X/D = 3,4,$and 5. Elevated turbulence levels are observed in the central wake region, where strong velocity fluctuations are induced by the vortex-shedding process. With increasing downstream distance, the turbulence distribution broadens and the peak values exhibit a slight decrease, reflecting the progressive dissipation of turbulent energy. This behavior is physically consistent, as vortex structures lose energy while being convected downstream.

Moreover, the symmetry of the profiles about $Z/D = 0$confirms that the measurement positioning was correctly aligned and that the wake is well centered relative to the cylinder axis.



The third graph illustrates the frequency spectrum of the velocity signal. A dominant peak is clearly visible in the spectrum, corresponding to the vortex-shedding frequency generated by the cylinder. The presence of this well-defined peak constitutes a key result, as it confirms that the flow is in a periodic shedding regime. The lower-amplitude background noise is associated with small-scale turbulence and electronic interference. The distinct separation between the dominant frequency and the noise demonstrates that the acquisition time and sampling parameters were appropriately selected.

This outcome is fully consistent with the classical von Kármán vortex street observed behind a circular cylinder, where periodic vortex shedding produces a characteristic frequency that dominates the spectral distribution.

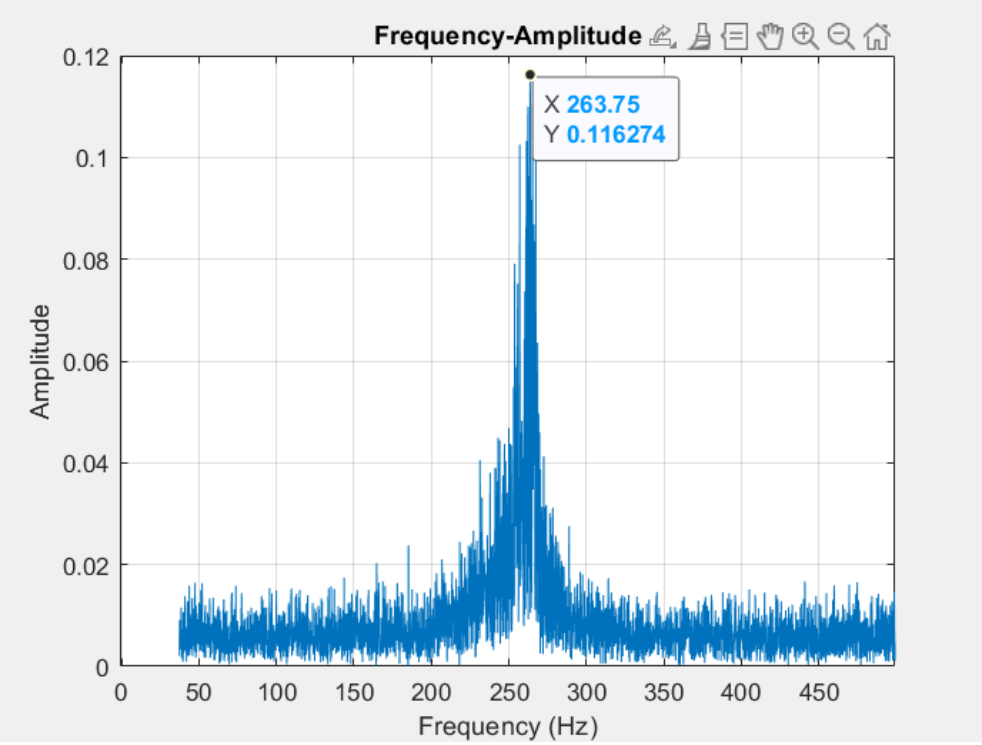


The fourth graph presents the Strouhal number as a function of Z/D for different downstream positions defined by $X/D = 3,4,$ and 5. The Strouhal number remains nearly constant across most of the wake for a given X/D, as well as between successive sections, with only minor variations observed near the wake edges. This result is particularly significant, since the Strouhal number is theoretically independent of the transverse position and depends primarily on the Reynolds number. The experimentally

obtained values fall within the expected range for a circular cylinder at the corresponding Reynolds number, thereby confirming the validity of both the frequency extraction and the velocity measurements. The slight dispersion observed at certain locations can be attributed to local turbulence effects, reduced signal-to-noise ratio, and edge influences.

Taken together, the four graphs demonstrate global coherence with the theoretical behavior of a cylinder wake. The velocity deficit, the turbulence distribution, the presence of a dominant shedding frequency, and the near constancy of the Strouhal number all provide strong evidence that the experiment and subsequent data processing were correctly executed. The MATLAB code played a central role in ensuring reliable conversion of raw voltage signals into physical quantities, performing spectral analysis, and enabling direct visualization of the flow behavior. These results validate the experimental methodology and establish consistency with theoretical predictions and reference data from the literature on wake flows behind bluff bodies.

Strouhal number



On this graph, the dominant peak is measured at
f = 263.75 Hz

The flow parameters remain unchanged.

$U\infty = 7.18$ m/s

$D = 0.006$ m

$v = 1.59 \times 10^{-5}$ m^2/s

Using only this frequency, the Strouhal number is calculated as follows.

$St = f * D / U\infty$

$St = 263.75 * 0.006 / 7.18$

$St = 0.220$

The Reynolds number remains unchanged since the frequency does not appear in its expression.

$Re = U\infty \times D / v$

$Re = 7.18 \times 0.006 / 1.59 \times 10^{-5}$

$Re = 2.71 \times 10^3$

With the first graph only, we therefore obtain.

$St = 0.220$

$Re = 2.71 \times 10^3$

When the experimental results are compared with the Strouhal–Reynolds reference curve provided in the course, a Reynolds number of approximately 2.7×10^3 corresponds to a theoretical Strouhal number of about 0.20 for a smooth circular cylinder. The experimentally obtained value of 0.220 is therefore in very close agreement with the theoretical prediction.

The dominant frequency identified in the first spectrum leads directly to a Strouhal number consistent with theory, confirming that the flow is in a periodic vortex-shedding regime. This outcome demonstrates the coherence and reliability of the entire measurement and processing chain, from hot-wire voltage acquisition to Fourier transform analysis and the determination of St and Re.

The small discrepancy between 0.20 and 0.220 can be attributed to normal experimental uncertainties, including velocity measurement accuracy, ambient temperature variations, and probe positioning. These differences remain fully acceptable and do not compromise the validity of the results.

Conclusion

This experimental study underscores the importance of rigorous experimental methods in fluid mechanics and demonstrates how physical measurements can be employed to describe complex flow phenomena. The wake behind a circular cylinder serves as a fundamental reference case, as it involves flow separation, turbulence, and periodic vortex shedding—key mechanisms in aerodynamics.

Throughout the investigation, the complete experimental process was addressed, from sensor calibration and data acquisition to signal processing and non-dimensional analysis. This workflow reflects real engineering practice, where raw measurements must be carefully converted into physical quantities before meaningful interpretation is possible. Moreover, the comparison of experimental results with theoretical models ensured both validity and physical coherence.

The controlled wind tunnel environment enabled a detailed study of the cylinder wake. Careful calibration of the hot-wire and pressure sensors provided reliable conversions from electrical signals to physical velocities. The spatial scanning of the wake, combined with extended acquisition times, allowed accurate determination of mean velocity, turbulence intensity, and dominant frequency. The MATLAB-based data processing ensured consistent treatment of all signals and facilitated the extraction of Strouhal and Reynolds numbers.

The results revealed a velocity deficit, turbulence distribution, and dominant vortex-shedding frequency that are all consistent with the theoretical behavior of a cylinder wake. The experimentally obtained Strouhal number was found to be very close to the theoretical reference value, thereby validating both the measurement methodology and the data processing chain.

Overall, the findings confirm the presence of a classical periodic von Kármán vortex street and demonstrate the reliability, accuracy, and physical consistency of the entire experimental procedure. This work highlights the essential role of experimental fluid mechanics in bridging theory and practice, and provides a robust framework for analyzing wake flows behind bluff bodies.

TP 4: Aerodynamic Drag Measurement

Introduction

The purpose of this study is twofold. First, it aims to develop an understanding of the calibration procedure for the strain gauge mounted in the wind tunnel. Accurate calibration is essential to ensure that the electrical signal delivered by the strain gauge can be reliably converted into a physical force measurement. Second, the experiment seeks to apply this calibrated system to the measurement of drag forces acting on bluff and streamlined bodies. Specifically, the drag force and corresponding drag coefficients are determined for a circular cylinder and a teardrop-shaped cylinder placed in a uniform airflow.

By combining careful calibration with systematic measurements, the experiment provides insight into the aerodynamic behavior of different geometries and highlights the influence of body shape on drag generation.

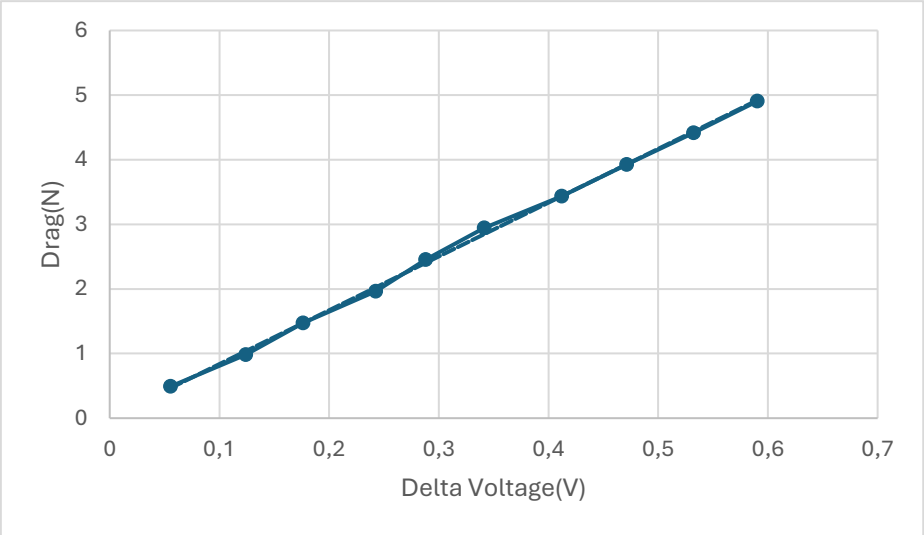





Experiment and results

The first step of the experiment consisted in calibrating the strain gauge of the wind tunnel balance. For this purpose, a series of known weights ranging from 200 g to 700 g, with increments of 50 g, were successively applied. For each load, the corresponding voltage output from the strain gauge was recorded.

This procedure allowed the establishment of a direct relationship between the applied drag force and the measured voltage. The paired values were organized in an Excel spreadsheet, and the resulting calibration curve was plotted. The graph clearly shows a linear trend, confirming that the strain gauge response is proportional to the applied force. This calibration law is essential, as it enables the conversion of voltage signals into drag forces during subsequent aerodynamic measurements.



Analysis

The calibration graph reveals a clear linear relationship between the measured voltage and the applied drag force. The coefficient of determination, R^2, is very close to unity, which demonstrates the robustness of the linear fit and confirms the reliability of the calibration procedure. This strong correlation indicates that the strain gauge response is highly consistent and that the conversion from voltage to force can be performed with confidence.

Such accuracy is crucial, as it ensures that all subsequent aerodynamic measurements—particularly the determination of drag forces and drag coefficients—are both precise and trustworthy. The calibration therefore constitutes a fundamental step in the experimental process, providing the necessary basis for quantitative analysis of the circular and teardrop cylinders.

Next step

Following the calibration of the strain gauge, a preload of 200 g was applied to the test model. This ensured that the system began with a known reference force, thereby establishing a baseline for subsequent wind tunnel measurements.

The wind tunnel was then operated over a range of excitation frequencies: from 15 Hz to 30 Hz in increments of 1 Hz, and from 30 Hz to 40 Hz in increments of 5 Hz. At each

operating point, the voltage output of the strain gauge was recorded, corresponding to the drag force experienced by the model.

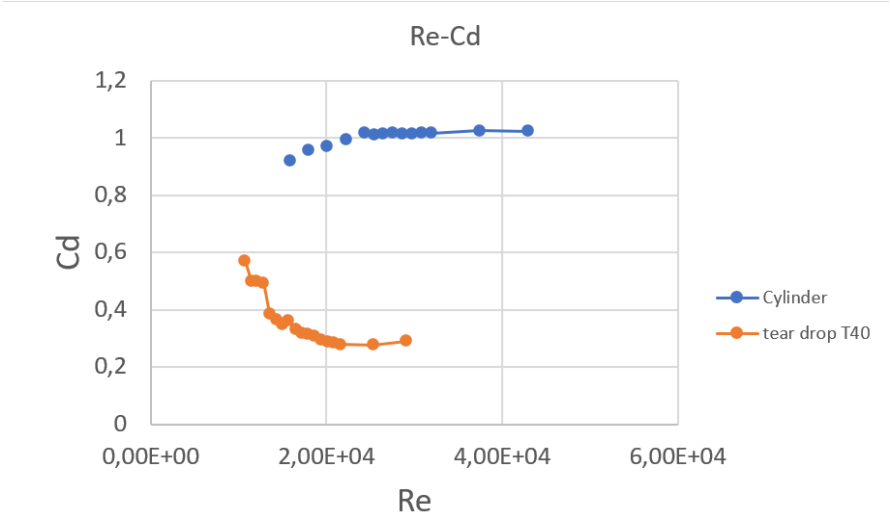
The experiment was repeated for two different geometries: first with the circular cylinder, and then with the teardrop-shaped cylinder. This comparative approach provided insight into how body shape and frontal area influence the drag force under identical flow conditions.

The recorded data were processed using Excel to establish the relationship between Reynolds number and drag coefficient. These two quantities were calculated using the following expressions:

$$C_D = \frac{2D}{\rho A V^2} \text{ and } Re = \frac{\rho V L}{\mu}$$

where $L$is the characteristic thickness of the model, $A$is the reference area, $\rho$is the air density, $V$is the free-stream velocity, and $\mu$is the dynamic viscosity of air.

The resulting relation between Reynolds number and drag coefficient is represented graphically. The plot highlights the aerodynamic differences between the circular and teardrop cylinders, illustrating the influence of geometry on drag behavior.



Analysis

The graphic compares the variation of drag coefficient (C_D) with Reynolds number (Re) for two geometries: a circular cylinder (blue curve) and a teardrop-shaped cylinder (orange curve). The circular cylinder consistently exhibits a higher drag coefficient across the full range of Reynolds numbers, reflecting its greater resistance to fluid flow due to pronounced flow separation and pressure drag.

In contrast, the teardrop shape demonstrates a marked decrease in $C_D$as Reincreases, highlighting its aerodynamic efficiency. This behavior confirms the design principle of streamlined bodies, which minimize drag by delaying flow separation and reducing turbulence in the wake.

Both curves also illustrate the transition from laminar to turbulent flow regimes. The circular cylinder shows a relatively stable high drag coefficient, while the teardrop curve emphasizes the benefits of streamlined geometry in reducing drag forces. The accompanying schematic representation of the laminar–turbulent transition further reinforces the physical interpretation of these results.

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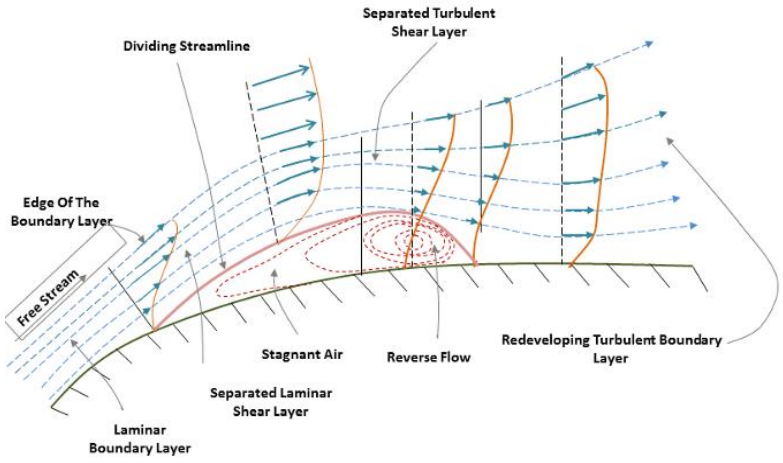


Fig 3, Laminar Separation Bubble [1]

It should be noted that the orange curve representing the teardrop cylinder is not perfectly smooth. This irregularity arises from the limited precision of the experimental setup, which cannot achieve absolute accuracy. Minor fluctuations in the data are expected due to measurement uncertainties, probe positioning, and flow variability. Nevertheless, the overall trend of the curve remains physically coherent, and the observed decrease in $C_D$with increasing Reclearly reflects the aerodynamic efficiency of the teardrop geometry.

Conclusion

This experiment provided valuable insight into the calibration and application of a strain gauge for aerodynamic testing. The calibration procedure established a reliable linear relationship between voltage and drag force, ensuring that subsequent measurements could be interpreted with precision.

Tests conducted on two different geometries—a circular cylinder and a teardrop-shaped cylinder—enabled the calculation of drag coefficients and Reynolds numbers across a range of flow conditions. The comparison highlighted the strong influence of body shape on aerodynamic performance: the circular cylinder consistently exhibited higher drag coefficients, while the teardrop geometry demonstrated significantly lower values, confirming its superior aerodynamic efficiency through reduced flow separation and turbulence.

The results also revealed the transition from laminar to turbulent flow regimes, further illustrating the complexity of wake behavior and the dependence of drag characteristics on Reynolds number. Together, these findings underscore the importance of careful calibration, systematic measurement, and comparative analysis in fluid mechanics.

Overall, the experiment not only validated the methodology but also contributed to a deeper understanding of fundamental aerodynamic principles, bridging theoretical models with practical observations.

The end